

Project Design Phase-II Technology Stack (Architecture & Stack)

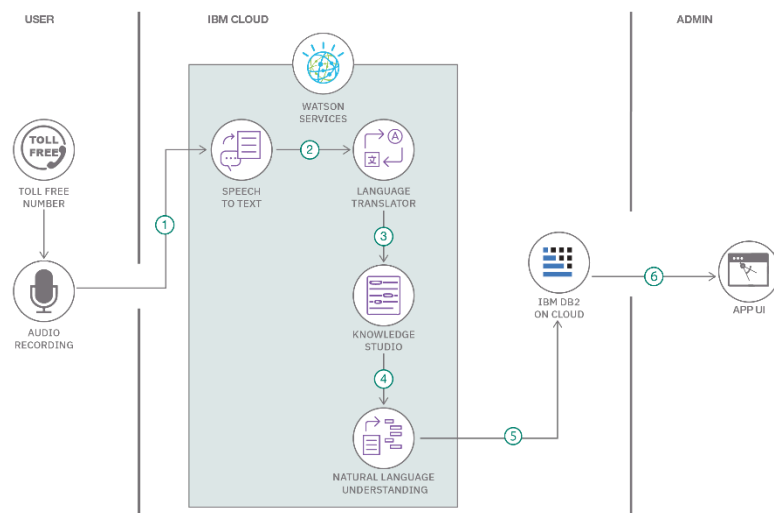
Date	27 JUNE 2026
Team ID	NIL/24761A05O7
Project Name	FOOD ORDERING AND CUSTOMER TRENDS WITHIN BUSINESS
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	WE DASHBOARD FOR MANAGERS TO VIEW ANALYTICS	TABLEAU PUBLIC OR DESKTOP POWER BI EXCEL
2.	Application Logic-1	DATA EXTRACTION FROM CSV	Python , EXCEL ,TABLEAU
3.	Application Logic-2	DATA CLEANING AND TRASFORMATION	NUMPY, TABLEAU, EXCEL, TABLEAU CALCULATED FIELDS, LOD EXPRESSIONS
4.	Application Logic-3	KPIS	PYTHON, TABLEAU
5.	Database	STORAGE	MySQL, BIG QUERY.
6.	Cloud Database	BACKUP	IBM DB2, IBM Cloudant , GOOGLE, AWS, RDS
7.	File Storage	RAW CSV UPLOADS FROM RESTAURANT	GOOGLE DRIVE
8.	External API-1	GET RESTAURANT DETAILS AND CUSINE	GOOGLE MAPS API
9.	External API-2	NOT NEEDED FOR BASIC DASHBOARD	Aadhar API, etc.
10.	Machine Learning Model	OPTIONAL PREDICT SALES TRENDS	SCIKIT-LEARN, LINEAR REGRESSION
11.	Infrastructure (Server / Cloud)	DEPLOY DASHBOARD+DB	Local, Cloud Foundry, Kubernetes, AWS..

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	TOOLS USED FOR ETL AND ANALYSIS	PYTHON , PANDAS AND NUMPY

S.No	Characteristics	Description	Technology
2.	Security Implementations	DATA ACCESS CONTROL	TABLEAU ROW-LVL SECURITY AND PASSWORD
3.	Scalable Architecture	3-TIER DATA SOURCE TO TABLEAU PREP-ETL-TABLEAU VIZ HANDLES MORE RESTAURANT ROWS	TABLEAU+MYSQL
4.	Availability	DASHBOARD ACCESS	TABLEAU PUBLIC
5.	Performance	FAST LOAD FOR LARGE DATASETS	TABLEAU EXTRACTS.HYPE,DATA SOURCE FILTERS

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>